

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

1. Star Topology vs Bus Topology

Cost

Bus topology is cheaper because it uses a single backbone cable and requires fewer devices. Star topology requires a central switch/hub and separate cables for every device.

Reliability

Star topology is more reliable. If one device or its cable fails, only that device is affected. In bus topology, failure of the main backbone cable can affect the entire network.

Performance

Star topology provides better performance because the central switch manages communication and reduces collisions. Bus topology can become slower when more devices are connected because all devices share the same communication medium.

Comparison

Factor	Star	Bus
Cost	Higher	Lower
Reliability	High	Low
Performance	Better	Decreases with traffic
Cabling	More	Less
Failure	One node affected	Main cable failure affects network

Conclusion

Star topology is better for reliability and performance, while bus topology is better when low cost is the main requirement.

2. Mesh Topology vs Star Topology – Fault Tolerance

Mesh Topology

In mesh topology, devices have multiple connections and alternative paths are available. If one link fails, data can travel through another path.

Star Topology

In star topology, each device connects to a central switch or hub. Failure of one individual cable affects only that device. However, failure of the central switch can stop the entire network.

Comparison

Factor	Mesh	Star
Fault tolerance	Very High	Moderate
Alternative paths	Multiple	Usually one central path
Reliability	Very High	High
Central device	Not required	Required
Cost	High	Moderate

Conclusion

Mesh topology is more fault-tolerant and reliable than star topology, making it suitable for critical networks where continuous communication is important.

3. Bus Topology vs Mesh Topology – Installation Complexity

Bus Topology

Bus topology uses a **single backbone cable**. Devices are connected to this common cable, so installation is simple and requires fewer cables and connections.

Mesh Topology

Mesh topology requires many connections between devices. As the number of devices increases, the number of cables and ports also increases.

Comparison

Factor	Bus	Mesh
Installation	Easy	Complex
Cabling	Less	More
Cost	Low	High
Configuration	Simple	Difficult
Scalability	Limited	Complex as devices increase

Conclusion

Bus topology is easier and cheaper to install, while mesh topology is more complex and expensive because it requires many connections.

4. Hybrid Topology vs Star Topology – Scalability

Hybrid Topology

Hybrid topology combines two or more topologies such as star, mesh, or bus. New sections can be added according to the organization's requirements.

Star Topology

Star topology can also be expanded, but its expansion depends on the available ports of the central switch or hub. Additional switches may be required when all ports are occupied.

Comparison

Factor	Hybrid	Star
Scalability	Very High	High
Expansion	Flexible	Depends on central device
Flexibility	High	Moderate
Additional hardware	As required	Required after port limit
Suitable for	Large networks	Small/medium networks

Conclusion

Hybrid topology provides greater scalability and flexibility than star topology, especially for large organizations with changing network requirements.

5. Maintenance Comparison of Star, Mesh, Bus and Hybrid Topologies

Star Topology

Star topology is easy to maintain because each device has a separate connection. A faulty device or cable can be identified and isolated easily.

Mesh Topology

Mesh topology is difficult to maintain because it contains many connections. Finding and troubleshooting faults can take more time.

Bus Topology

Bus topology is harder to maintain because a problem in the main backbone cable can affect the entire network, and locating the fault can be difficult.

Hybrid Topology

Hybrid topology has moderate maintenance complexity. It provides flexibility, but managing multiple types of topologies requires proper monitoring and skilled administration.

Comparison

Topology	Maintenance	Main Reason
Star	Easy	Faults are easy to isolate
Mesh	Difficult	Many connections
Bus	Difficult	Backbone failure affects network
Hybrid	Moderate	Multiple topologies must be managed

Conclusion

Star topology is the easiest to maintain, while mesh topology is the most complex. Bus topology has difficult backbone troubleshooting, and hybrid topology requires skilled administration because it combines different network structures.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
		Uses strong,			

Use of Examples / Evidence	10%	relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion